Pleural Diseases

Anatomy and Physiology of the pleura:

The pleura is a serous membrane covering the lungs, mediastinum, diaphragm, and inner chest wall. It consists of the visceral pleura, which envelops the lung surface and interlobar fissures, and the parietal pleura, which lines the thoracic cavity and is subdivided into costal, mediastinal, and diaphragmatic components. Both layers contain mesothelial cells that secrete lubricating fluid to allow smooth respiratory movement. The visceral and parietal pleura meet at the lung root, where the mediastinal pleura reflects laterally over the hilum and forms the pulmonary ligament posteriorly.

The narrow pleural space between the two layers contains a small amount of fluid and is completely separated between right and left by the mediastinum. Blood supply is primarily from the intercostal arteries, with venous drainage to the chest wall veins. The parietal pleura is richly innervated by intercostal nerves, producing somatic pain with trauma or tumor invasion, whereas the visceral, mediastinal, and diaphragmatic pleura are supplied mainly by the phrenic nerve.

Normal pleural fluid volume is approximately 0.25 mL/kg, with a turnover of 0.15 mL/kg/hour. Fluid dynamics are governed by Starling forces, capillary permeability, lymphatic drainage, and negative intrathoracic pressure. Fluid is filtered mainly from parietal pleural capillaries and reabsorbed by parietal lymphatics. Disruption of these mechanisms leads to pleural effusion. The pleural fluid's main functions are to reduce friction, maintain lung–chest wall apposition, and ensure effective mechanical coupling during breathing.

Common pleural disorders

1. Pneumothorax
2. Hydrothorax including empyema thoracis
3. Hydropneumothorax
4. Hemothorax is discussed in thoracic trauma lecture
5. Tumors

Pneumothorax

Overview

• Abnormal presence of air within the pleural cavity
• Results in dissociation of the parietal and visceral pleura leading to a disruption of lung mechanics.
• Lung compression reduces lung compliance, volumes, and diffusion capacity and causes ventilation-perfusion mismatch and poor gas exchange

Types of pneumothoraxes

1. **Spontaneous:** Primary spontaneous pneumothorax
2. **Secondary spontaneous pneumothorax**; due to underlying lung disease such as: Chronic obstructive pulmonary disease, Bronchiolitis, Pneumocystis infection, Cystic fibrosis, Asthma, Necrotizing pneumonia, Tuberculosis infection, Congenital cysts, Pulmonary fibrosis, Catamenial
3. **Trauma Blunt** thoracic injury**,** Penetrating injury
4. **Iatrogenic** as in Barotrauma of mechanical ventilation, Percutaneous transthoracic lung biopsy, Central venous catheterization
5. **Thoracotomy/thoracostomy**
6. **Others**, such as in Bronchopleural fistula, Esophageal perforation

Tension Pneumothorax

• If pneumothorax is left untreated, air can accumulate without decompressing adequately, leading to high positive pleural pressures, causing severe lung collapse, compression of the mediastinum with deviation to the contralateral side causing great vessels and heart compression, and ultimately hemodynamic compromise.

• Patients in tension pneumothorax require immediate clinical diagnosis, followed by needle decompression in the second intercostal space at the midclavicular line, followed by tube thoracostomy.

Etiology of Pneumothorax

Primary Spontaneous Pneumothorax (PSP): no underlying trauma or iatrogenic cause for p pneumothorax could be identified, Incidence is 7.4/100,000 (male) and 1.2/100,000 (females annually. PSP is caused by the rupture of small blebs (bullae) usually in apices of the upper or lower lobes, allowing air to leak into the pleural cavity. About 80 % of patients will demonstrate emphysema-like changes on CT scan. PSP occurs most commonly in thin tall young male patients. Smoking and atmospheric pressure changes are also risk factors.

Secondary Spontaneous Pneumothorax*:* known underlying lung disease. Incidence is 6.3/100,000 (men) and 2.0/100,000 (women) annually. Underlying disease causing lung parenchymal rupture and pneumothorax.

Clinical Presentation

Sudden onset ipsilateral pleuritic chest pain and dyspnea (most common), associated with tachypnoea, and tachycardia.

Physical examination: may be normal if the pneumothorax is small i.e. <25 % of the lung is collapsed. Decreased breath sounds, and hyperresonance on the affected side (not always present), Subcutaneous emphysema may present in traumatic pneumothorax

Tension pneumothorax: cases would present with signs of respiratory distress (severe dyspnea, peripheral and/or central cyanosis, tachypnea, distended neck veins, tachycardia, hypotension also presents with tracheal deviation to the contralateral side Rarely: pneumomediastinum or pneumopericardium(Hamman's sign).

Diagnosis:

This is established by an upright chest X-ray. If clinical signs of tension pneumothorax is present chest X-ray confirmation should be omitted and immediate decompression should ensue. Pneumothorax in supine patients accumulates into the dependent regions of the anterior and subdiaphragmatic pleura and may be detected as a deep sulcus sign.

Although CT scan is the gold-standard for diagnosis, it is not necessary for the majority of patients with first episode spontaneous pneumothorax since it does not change the management. Ultrasound is also an accurate, rapid and non-invasive test, yet requires operator experience.

Management

• The etiology largely dictates both immediate and definitive management.

Conservative Observation mainly in PSP: Reserved for asymptomatic patients with a small pneumothorax who are unlikely to have an ongoing air leak. Follow-up radiography should be obtained within 24–48 hours to document improvement. Supplemental oxygen can help decrease the alveolar pressure of nitrogen in the body, thus creating a gradient to reabsorb the air from the pleura (mostly composed of nitrogen) into the alveoli and tissues. Conservative management re-expands the lung at an average rate of 2.2 % / day with a 79 % success rate. Failure for the pneumothorax to resolve in 48 hours needs intervention.

• Aspiration: A small cannula can be used to aspirate pleural air, with a success rate of 75 and 40 % for a primary and secondary spontaneous pneumothorax, respectively. This technique is rarely used. Can be attached to a Heimlich one-way valve or a three-way stop-cock with a large syringe.

• Percutaneous Catheters: Small-caliber tube thoracostomy can be performed percutaneously via the Seldinger technique and attached to either a Heimlich one-way valve or suction. Use is limited to a spontaneous pneumothorax with limited respiratory symptoms.

• Tube Thoracostomy: Large-bore chest tubes are the standard of care for the treatment of traumatic pneumothoraces, unstable patients, persistent or large air leaks and associated effusions or haemothorax.

Recurrence after spontaneous pneumothorax is 30 % after the first episode, with the majority occurring within 2 years . Independent risk factors for recurrence include: pulmonary fibrosis, age >60 years, increased height/weight rati]. Recurrence rate increases after each episode.

• Surgery: Guidelines recommend waiting at least 3–5 days for resolution of a spontaneous pneumothorax before considering definitive surgical management. A bullectomy can be performed preferably via a VATS approach with a success rate >95 %. Other options including open thoracotomy and axillary approaches.

Indications for Operative Intervention:

1. Persistent air leak (> 5 days) or failure of lung to fully re-expand with chest tube drainage
2. Hemopneumothorax
3. Recurrent ipsilateral pneumothorax
4. Bilateral spontaneous pneumothorax
5. First occurrence of contralateral pneumothorax
6. At-risk activities/professions (i.e., pilots, divers)
7. Poor access to medical treatment or follow-up

•Pleurodesis: Although surgery as definitive management of recurrent secondary pneumothoraces is preferred, pleurodesis with talc, bleomycin or doxycycline remains an option.

Pleural Effusions

Is an abnormal collection of fluid in the pleural space Can be transudative (low protein) or exudative (high protein). Benign pleural effusions are twice as common as malignant pleural effusions.

Empyema Thoracis

DEFINITIONS *Empyema*, from the Greek, is defined simply as “pus in the pleural cavity.” The precursor of empyema is bacterial pneumonia and subsequent parapneumonic effusion.

Causes:
1. Pulmonary infections: Complicated and unresolved pneumonia, Bronchiectasis, Tuberculosis, Lung abscess, intra-pleural ruptured hydatid cyst

2. Aspiration of pleural effusion for any cause may end by empyema if infected

3. Trauma: Blunt and penetrating chest trauma, thoracic surgery, esophageal surgery ,

4. Extra pulmonary causes: Subphrenic abscess

Clinical features:
There are symptoms of pus at any site like swinging pyrexia with generalize malaise, purulent productive cough, chest pain. Parapneumonic effusions occur in 20–60% of patients hospitalized for bacterial pneumonia; 5–10% of these parapneumonic effusions progress to empyema. The mortality rate of empyema is significant: 25–75% in the elderly and debilitated.
Parapneumonic effusions are exudative and progress through three stages to an empyema:
1. *First* or *exudative stage*, characterized by relatively low LDH, normal glucose, and normal pH of the pleural filud.
2. *Second* or *fibropurulent stage* evolves with invasion of pleural fluid by bacteria, increased fibrin deposition, cellular debris, and white blood cells with the ultimate formation of limiting fibrin membranes producing loculations
3. *Third* or *organization stage* occurs as fibroblasts grow into the exudative fibrin sheet coating the visceral and parietal pleura with an inelastic membrane or pleural "peel" encasing the lung and rendering it functionless and it's the most difficult stage for treatment.

Investigations:
1.Haematological Leukocytosis, elevated C. reactive protein and ESR
2.Standard posterior-anterior and lateral chest radiography shows pleural effusion, hydropneumothorax, fibrothorax in late stages
3. Pleural fluid first must be determined as exudative or transudative, which is accomplished by examination of pleural fluid obtained by thoracentesis criteria based on levels of lactate dehydrogenase (LDH) and protein concentrations: an exudate as having any one of the following characteristics:

A. pleural fluid protein divided by serum protein concentration greater than 0.5,
B. pleural fluid LDH concentration divided by serum LDH concentration greater than 0.6,
C. pleural fluid LDH concentration greater than two thirds of the upper limit of normal serum LDH concentration, and
D. A pH less than 7.0.

4. Ultrasonography is widely available in most institutions and frequently employed after the initial chest X-ray. Ultrasound is rapid, portable, and less expensive than a chest CT scan which is some time is necessary to identify the loculation.

5) Chest CT scan

Empyema is determined by analysis of the pleural fluid, and treatment is guided by radiological assessment

MEDICAL MANAGEMENT

Medical management or conservative noninterventional therapy is rarely effective and often contraindicated for management of empyema. Thoracentesis and culture sensitivity–based antibiotic therapy are appropriate and generally successful for stage I parapneumonic effusions but not stage II effusions or stage III empyema.
Drainage of the pleural space by radiologically guided catheters or surgical drainage must be employed for successful management of empyema. Tube thoracostomy using a large-bore chest tube (32–38 Fr) was the initial intervention when empyema was established.

After chest tube placement, fibrinolytics are administered until the pleural space is cleared radiographically and the patient's clinical condition is improved. Streptokinase was initially used. Subsequently, urokinase was used and found to be more efficacious.

SURGICAL MANAGEMENT

About 20–30% require a surgical drainage procedure:
I) Thoracoscopy and debridement should be the next therapeutic maneuver after attempted fibrinolysis. VATS must accomplish two therapeutic goals to be successful: (1) establish a unified pleural space and (2) ensure total reexpansion of the lung parenchyma with obliteration of the pleural cavity
II) Decortication via a thoracotomy should be performed when the third or fibrotic stage of empyema is suggested by CT scan that reveals visceral pleural enhancement without fibrin septation in multiple areas of loculation. *Decortication* refers to the process of peeling this restrictive fibrous layer from the pleura, promoting lung reexpansion and improving thoracic excursion
III) Open widow thoracostomy sometimes required

Malignant Pleural Effusions

Up to 22% of pleural effusions are caused by malignant disease. Affected patients with advanced neoplastic diseases experience considerable morbidity as a result of these pleural fluid collections.

ETIOLOGY

This may be due to primary or secondary tumors of the pleura with seeding of the intrapleural space and lymphatic obstruction. Of pleural neoplasms, 95% arise from a metastatic source, with lung and breast carcinoma accounting for 75% of all cases. Other common causes are lymphoma, gastric cancer, and ovarian cancer. Adenocarcinomas of unknown primary constitute a distinct entity in patients in whom the source of the pleural malignancy is never found. They are associated with exposure to environmental tobacco smoke.

Clinical features and Physical Findings:
Patients with malignant pleural effusions are typically symptomatic and complain of dyspnea, cough, or chest pain. The discomfort often is independent of respiration and but is aggravated by activity. With invasive pleural metastases, the affected nerve roots may radiate pain.

Physical findings indicating pleural effusion are decreased tactile fremitus and dullness to percussion on examination of the posterior chest.

Imaging

1) Early evidence of pleural effusion by chest radiograph is costo-diaphragmatic sulcus blunting caused by as little as 125 ml of fluid.
2) Computed tomography (CT) multiple pleural nodules and nodular pleural thickening are generally limited to effusions of malignant etiology.
3)Thoracic ultrasound shows the optimal site for diagnostic thoracentesis of a small pleural effusion. It also helps select ideal entry points for thoracoscopes or pleural biopsy instruments by avoiding adhesions.

Diagnostic Procedures

The diagnosis of pleural effusion is confirmed by thoracentesis. Practically all malignant pleural effusions are exudates. Criteria sets to classify effusions as exudative

Pleural Cytology and Biopsy

Cytological evaluations with standard staining (Papanicolaou's smears) generally confirm malignant pleural effusions. When standard pleural effusion cytology is nondiagnostic, pleural needle biopsy is indicated

TREATMENT

The optimal treatment for malignant pleural effusion is controversial. The best goal to achieve is drainage of the fluid and prevention of recurrence by Chemical pleurodesis by talc powder or chemotherapy drugs like Bleomycien or By irritant agents like Tetracycline .

Chylothorax

Definition: Chylothorax is the presence of lymph fluid in the pleural space.

Chylothorax may be iatrogenic, traumatic, or spontaneous because of congenital or acquired disorders. Early recognition is crucial to preventing serious complications of chronic lymphatic loss. Initial drainage, combined with measures to reduce chyle flow and maintain nutrition, may resolve the condition. For cases that do not show prompt response, particularly in the postoperative or trauma setting, early surgical ligation of the thoracic duct should be performed with the expectation of prompt resolution. Thoracoscopy offers a minimally invasive approach and warrants consideration by experienced surgeons. Other alternatives require consideration of the underlying etiology, consideration of the patient's medical condition and prognosis, and an understanding of the limitations of these approaches.

Tumors of the Pleura: Malignant Pleural:

Primary tumors of the pleura are very rare. There are two main types:

1. **Malignant Pleural Mesothelioma (MPM):** most common primary tumor of the pleura. Highly aggressive tumor with grave prognosis.
2. **Localized Fibrous Tumor of the Pleura:** significantly less aggressive than MPM, also referred to as "localized mesothelioma."

Epidemiology: Strong association with asbestos exposure (>80 % o cases), with 20-years latency until the development of MM. 5:1 Male predominance, with peak incidence in the sixth and seventh decade. Other causes: radiation, mineral fibers, simian virus 40.

Clinical Presentation

- Dyspnea
- Pain secondary to chest wall invasion and involvement of somatic nerves of the parietal pleura
- Malignant pleural effusion
- Pleural thickening, subtle pleural masses on radiographic images, Thick, restrictive pleural rind (late finding) with encasement of the lung

Work-Up

- CT chest with IV contrast +/− FDG-PET

- MRI as an adjunct to determine depth of chest wall invasion
- or diaphragmatic involvement
- Thoracocentesis of malignant effusion: cytology and elevated pleural hyaluronic acid have a 50 % diagnostic accuracy.
- Gold standard for diagnosis is thoracoscopic pleural biopsy of about 80 % Diagnostic accuracy

Management

• Most patients present with locally advanced disease and not amenable to surgical treatment options.

• Surgery: *Extra pleural Pneumonectomy (EPP):* en-bloc resection of the lung, visceral pleura, parietal pleura, pericardium, hemidiaphragm. Surgery is reserved for patients with localized and selected locally advanced disease. Surgery is provided in the context of multi-modality therapy (neoadjuvant or adjuvant chemotherapy and adjuvant whole thorax radiation therapy), with improvement in overall survival and disease-free survival depending on tumor histopathology and regimen.

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